

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 2, Double-blind, Randomized, Placebo-controlled Study to Evaluate the Effects of Sotatercept Versus Placebo for the Treatment of Combined Postcapillary and Precapillary Pulmonary Hypertension (Cpc-PH) Due to Heart Failure With Preserved Ejection Fraction (HFpEF)
Short Title:	A Study of Sotatercept for the Treatment of Cpc-PH Due to HFpEF (MK-7962-007/A011-16)
Protocol Number:	339426069472573451
NCT Number:	NCT04945460
Sponsor Name:	Merck
Investigational Product:	Sotatercept
Phase:	PHASE2
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Change From Baseline in Pulmonary Vascular Resistance (PVR) at Week 24: PVR, a hemodynamic variable of pulmonary circulation, is measured by right heart catheterization (RHC).
Secondary Obj.:	1. Change From Baseline in the 6-Minute Walk Distance (6MWD) at Week 24 2. Time to First Clinical Worsening Event (TTCW) at Weeks 24 3. Change From Baseline in Mean Pulmonary Arterial Pressure (mPAP) at Week 24

2.2 Background & Rationale

This is a Phase 2, double-blind, randomized, placebo-controlled study to evaluate the efficacy and safety of sotatercept versus placebo in adults with Cpc-PH due to HFpEF.

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE2, Status: ACTIVE_NOT_RECRUITING
Target N:	164

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

Participants must meet the following criteria to be enrolled in this proof-of-concept study:

- Age 18 to 85 years
- Clinical diagnosis of HFpEF:

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? Left ventricular ejection fraction ?50%, with no history of LVEF below 45% in more than two consecutive measurements under stable conditions

3. Demonstrated Cpc-PH by all of the following:

- * Baseline RHC performed within 28 days of randomization documenting a minimum PVR of ?320 dyn?sec/cm⁵ (4 wood units)

- * Mean pulmonary arterial pressure (mPAP) of \>20 mmHg

- * Pulmonary capillary wedge pressure (PCWP) \>15 mmHg but \< 30 mmHg

4. New York Heart Association FC of II or III

5. Six-minute Walk Distance ?100 m repeated twice during Screening and both values within 15% of each other, calculated from the highest value

6. Chronic medication for HF or for any underlying condition, administered at a stable (per investigator) dose for ?30 days prior to Visit 1. Diuretics and/or anticoagulants are excepted from this rule but should not be newly started or stopped within 30 days of Visit 1, and a prescribed dose change should not occur within 7 days of Visit 1. Anticoagulation may be suspended for RHC if necessary.

7. Women of childbearing potential must:

- * Have 2 negative urine or serum pregnancy tests as verified by the investigator during the Screening Period; must agree to ongoing pregnancy testing during the course of the study and until 8 weeks after the last dose of the study drug

- * If sexually active with a male partner: use highly effective contraception without interruption for at least 28 days prior to starting the investigational product AND agree to use the same highly effective contraception in combination with a barrier method during the study (including dose interruptions), and for 16 weeks (112 days) after discontinuation of study treatment

- * Refrain from breastfeeding a child or donating blood, eggs, or ovum for the duration of the study and for at least 16 weeks (112 days) after the last dose of study drug

8. Male participants must:

- * Agree to use a condom, defined as a male latex condom or non latex condom NOT made out of natural (animal) membrane (e.g., polyurethane), during sexual contact with a pregnant female or a female of childbearing potential while participating in the study, during dose interruptions, and for at least 16 weeks (112 days) following investigational product discontinuation, even if he has undergone a successful vasectomy

- * Refrain from donating blood or sperm for the duration of the study and for 16 weeks (112 days) after the last dose of study drug

9. Ability to adhere to the study visit schedule and understand and comply with all protocol requirements

10. Agreement to not participate in any other trials of investigational drugs/devices while enrolled in the A011-16 study

11. Ability to understand and provide documented consent for participation

4.2 Exclusion Criteria

Participants will be excluded from the study if any of the following criteria are met:

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1. A diagnosis of PH in WHO Group 1, WHO Group 3, WHO Group 4, or WHO Group 5
2. Clinically significant and active lung disease:

- * Chronic obstructive pulmonary disease with post-bronchodilator forced expiratory volume in the first second (FEV1) $\leq 60\%$ predicted

- * Restrictive lung disease with total lung capacity $\leq 70\%$ predicted

- * More than mild interstitial lung disease (ILD), with FVC $\leq 70\%$ or FEV1 $\leq 60\%$ predicted (still appropriate if absence of more than mild ILD, fibrosis, or COPD on computed tomography [CT] imaging)

3. Cardiovascular co-morbidities, which include any of the following:

- * History of more than mild mitral or aortic stenosis

- * Ongoing more than mild mitral or aortic regurgitation

- * More than one valve replacement or repair (mechanical or biomechanical) or anticipation of any valve replacement or repair

- * Severe tricuspid regurgitation due to primary valvular disease

- * Occurrence of myocardial infarction, acute coronary syndrome, coronary artery bypass graft or percutaneous coronary intervention within 180 days of Visit 1

- * History of serious life-threatening or hemodynamically significant arrhythmia

- * History of or anticipated heart transplant or ventricular assist device implantation

- * History of implantable cardioverter defibrillator placement or anticipated implantation of pacemaker, pacemaker implantation within 30 days of Screening

- * Anticipated implantation of pacemaker, pacemaker implantation within 30 days of Screening or history of implantable cardioverter defibrillator placement

- * Occurrence of myocardial infarction within 180 days of Visit 1

- * History of known pericardial constriction, hypertrophic cardiomyopathy, sarcoidosis, or amyloid cardiomyopathy

- * Uncontrolled systemic hypertension as evidenced by sitting systolic blood pressure ≥ 160 mmHg or sitting diastolic blood pressure ≥ 110 mmHg during Screening after a period of rest

- * Systemic hypotension as evidenced by sitting systolic blood pressure ≤ 90 mmHg or sitting diastolic blood pressure ≤ 50 mmHg during Screening

- * Resting heart rate of ≤ 45 bpm or ≥ 115 bpm

- * Stroke within 90 days of Visit 1

- * Acutely decompensated HF that required hospitalization within 30 days of Visit 1

- * Electrocardiogram during Screening Period with Fridericia's corrected QT interval (QTcF) ≥ 470 msec for males or ≥ 480 msec for females, or ≥ 500 msec if a ventricular conduction defect (right bundle branch block; left bundle branch block; or interventricular conduction delay) is present

- * Personal or family history of Brugada syndrome, sudden cardiac arrest or unexplained sudden cardiac death or arrest

- * Personal or family history of long QT syndrome unless the participant's ECG shows a normal QTc

- * Arrhythmogenic right ventricular dysplasia (ARVD) unless the participant has a recent cardiac MRI that shows no evidence of this diagnosis

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4. Hospitalization for any worsening of medical conditions or any significant surgery per investigator within 30 days of Visit 1.
5. Received any approved PAH-specific therapies (i.e., endothelin receptor antagonists, prostacyclin analogs, phosphodiesterase-5 inhibitors, soluble guanylate cyclase stimulators) within 30 days of Visit 1. The use of an oral phosphodiesterase type 5 inhibitor, if only indicated for erectile dysfunction, is permitted, if not administered within 48 hours of a study visit or procedure.
6. Received intravenous inotropes (e.g., dobutamine, dopamine, norepinephrine, vasopressin, or levosimendan) within 30 days of Visit 1
7. Received erythropoietin within 6 months of Visit 1.
8. Known history of chronic liver disease, including untreated hepatitis B and/or hepatitis C (with evidence of recent infection and/or active virus replication), with severe hepatic impairment and/or cirrhosis (e.g., hepatic encephalopathy)
9. Prior exposure to sotatercept or luspatercept.
10. Currently enrolled in or have completed any other investigational product study within 30 days for small molecule drugs or within 5 half-lives for investigational biologics prior to the date of documented consent.
11. Initiation of an exercise program for cardiopulmonary rehabilitation within 90 days of Visit 1 or planned initiation during the study (participants who are stable in the maintenance phase of a program and who will continue for the duration of the study are eligible).
12. Any of the following clinical laboratory values prior to Visit 1 as specified:
 - * Hemoglobin (Hgb) above the gender-specific upper limit of normal (ULN) per local laboratory test within 28 days of Visit 1 or <10 g/dL per local laboratory within 28 days of Visit 1
 - * Serum alanine aminotransferase or aspartate aminotransferase levels $>3\times$ ULN or total bilirubin $>3\times$ ULN within 28 days of Visit 1
 - * Estimated glomerular filtration rate <30 ml/min/1.73 m² (4-variable Modification of Diet in Renal Disease equation) within 28 days of Visit 1 or required renal replacement therapy within 90 days of Visit 1
 - * Glycated hemoglobin (HbA1c) $>10\%$ within 28 days of Visit 1
 - * Platelet count $<75,000$ /mm³ within 28 days of Visit 1
13. History of severe allergic or anaphylactic reaction or hypersensitivity to recombinant proteins or excipients in the investigational product .
14. Major surgery within 60 days of Visit 1. Participants must have completely recovered from any previous surgery prior to Visit 1.
15. Prior organ transplantation (e.g., heart, lung, liver, kidney), bone marrow transplantation, or life expectancy of <12 months.
16. Pregnancy or breastfeeding in females.
17. Active malignancy, with the exception of fully excised or treated basal cell carcinoma, cervical carcinoma in situ, or ≥ 2 squamous cell carcinomas of the skin.
18. History of clinically significant (as determined by the investigator) endocrine, hematologic, hepatic, (auto)immune, infectious (requiring chronic antibiotics), metabolic, urologic, pulmonary, neurologic, neuromuscular, dermatologic, psychiatric, renal, and/or another disease that may limit participation in the study.
19. Body mass index ≥ 50 kg/m².

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20. More than mild obstructive sleep apnea (treated or untreated).

21. Any non-cardiopulmonary condition or acute/chronic impairment(s) (other than dyspnea) that limits the ability to perform 6-minute walk test (6MWT).

11. Study Identifiers & Governance

Primary ID: 339426069472573451

Registry Number: NCT04945460

Sponsor: Merck

Study Title: A Study of Sotatercept for the Treatment of Cpc-PH Due to HFpEF (MK-7962-007/A011-16)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____